Ch 8.2.1, 8.2.2: Bagging and Random Forests

Lecture 25 - CMSE 381

Prof. Guanqun Cao

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Mon, Nov 3, 2025

Announcements

Last time:

- 8.1 Decision Trees - regression

This lecture:

- 8.1 Decision Trees - classification
- 8.2.1 Bagging
- 8.2.2 Random forest

Announcements:

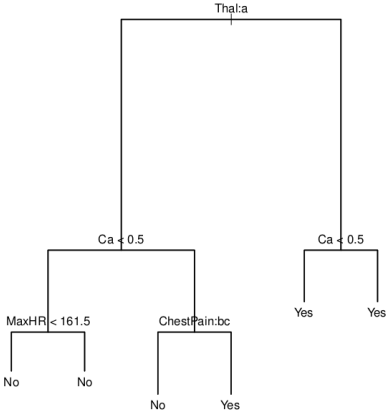
- Homework 7 Due Sunday

	F	10/17	***Review***		
	M	10/20	Fall Break		
	W	10/22	**Midterm #2**		
21	F	10/24	Polynomial & Step Functions	7.1-7.2	HW #5 Due Sun 10/28
22	M	10/27	Step Functions; Basis functions; Start Splines	7.2-7.4	
23	W	10/29	Regression Splines	7.4	
24	F	10/31	Decision Trees	8.1	HW #6 Due Sun 11/2
25	M	11/3	Random Forests	8.2.1, 8.2.2	
26	W	11/5	Maximal Margin Classifier	9.1	
27	F	11/7	SVC	9.2	HW #7 Due Sun 11/9
28	M	11/10	SVM	9.3, 9.4	
29	W	11/12	Single Layer NN	10.1	
30	F	11/13	Multi Layer NN	10.2	HW #8 Due Sun 11/16
31	M	11/17	CNN	10.3	
32	W	11/19	Unsupervised learning / clustering	12.1, 12.4	
33	F	11/21	Virtual: Project Office Hours		HW #9 Due Sun 11/23
	M	11/24	***Review***		
	W	11/26	**Midterm #3**		
	F	11/28	Thanksgiving		
	M	12/1	Virtual: Project Office Hours		
	W	12/3	Virtual: Project Office Hours		
	F	12/5			**Project Due**

Section 1

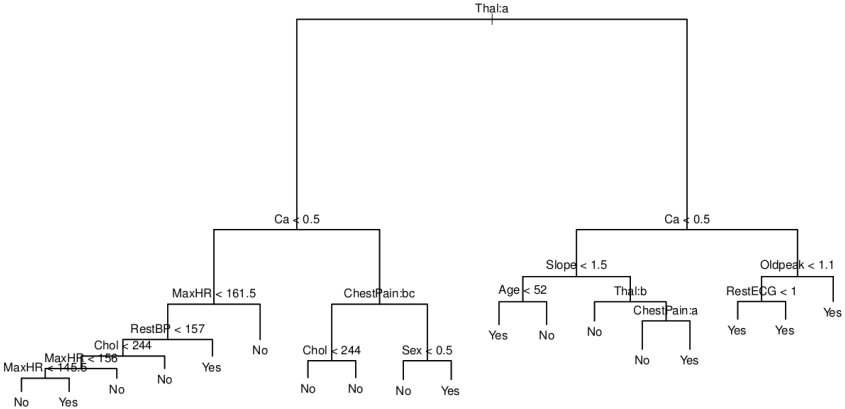
Classification Decision Tree

Basic idea



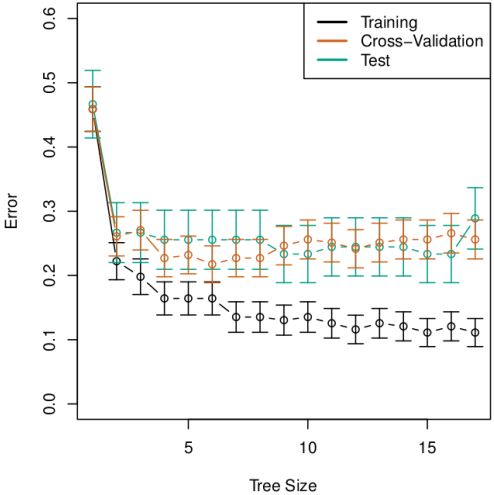
- $\hat{p}_{mk}$ = proportion of training observations in R_m from the kth class
- $E = 1 - \max_k(\hat{p}_{mk})$ *Fraction of training observations not in the most common class*
- Example from heart data
- Can't use RSS, need error rate
- Could decide splits by classification error rate
- Not necessarily the best, you can use other options but I'm going to skip that.

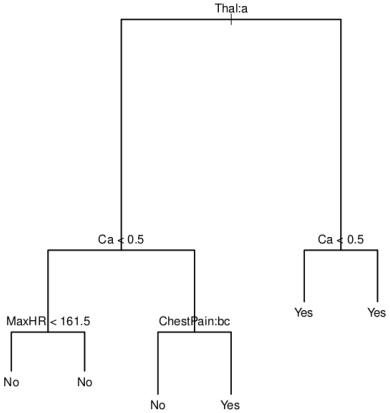
Example



The terminal nodes here: Yes = person has heart disease, No = no heart disease

Pruning the example

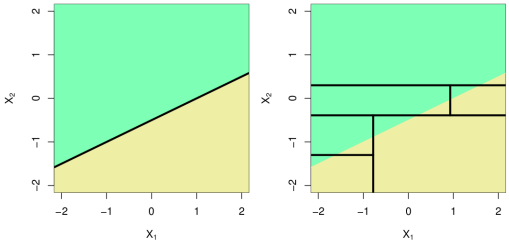




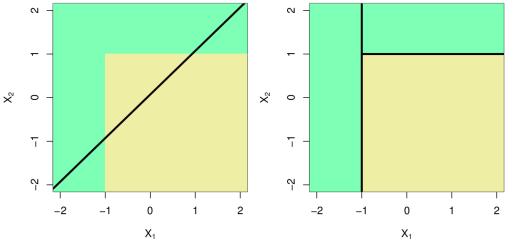
Coding!

Second part of day 24's jupyter notebook.

Linear models vs trees



Obviously linear does better here



Not going to beat this case though

Pros/Cons

Pros:

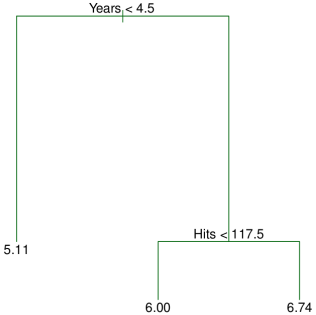
- Trees are very easy to explain to people. Often easier to explain than linear regression!
- Trees can be displayed graphically, and are easily interpreted even by a non-expert (especially if they are small).
- Trees can easily handle qualitative predictors without the need to create dummy variables.

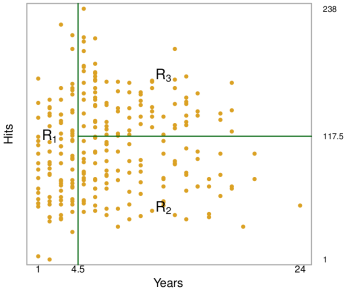
Cons:

- Not as accurate as other methods of classification and regression
- Not robust: small change in data can cause large change in estimated tree
- Fix..... aggregate many decision trees

TL;DR

- Split into regions by greedily decreasing RSS (or error rate)
- Prune tree by using cost complexity
- Not robust - Next, figure out how to aggregate trees





Section 2

8.2.1 Bagging

The bootstrap

Want to do (but can't):
Build separate models from independent training sets, and average resulting predictions:

- $\hat{f}^1(x), \cdots, \hat{f}^B(x)$ for B separate training sets
- Return the average

$$\hat{f}_{avg}(x) = \frac{1}{B}\sum_{b=1}^{B} \hat{f}^b(x)$$

- Decreases variance, but not practical

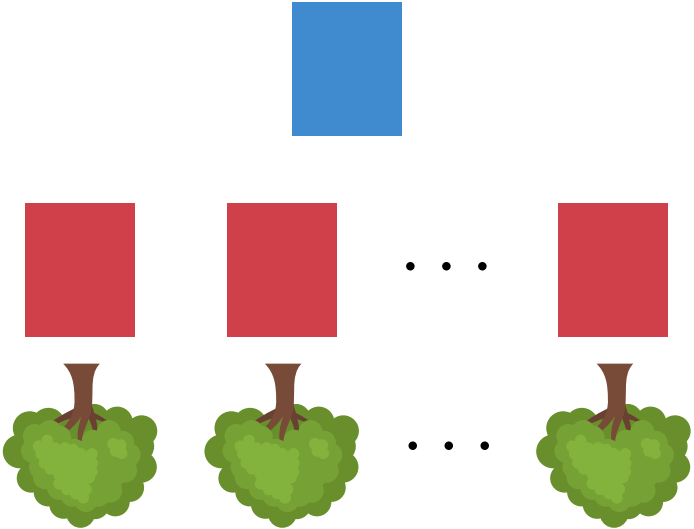
Boostrap modification:

- Work with fixed data set
- Take B samples from this data set (with replacement)
- Train method on bth sample to get $\hat{f}^{*b}(x)$
- Return average of predictions (regression)

$$\hat{f}_{bag}(x) = \frac{1}{B}\sum_{b=1}^{B} \hat{f}^{*b}(x)$$

or majority vote (classification)

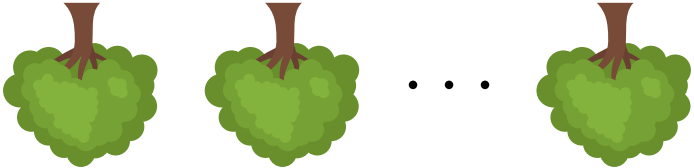
Tree version



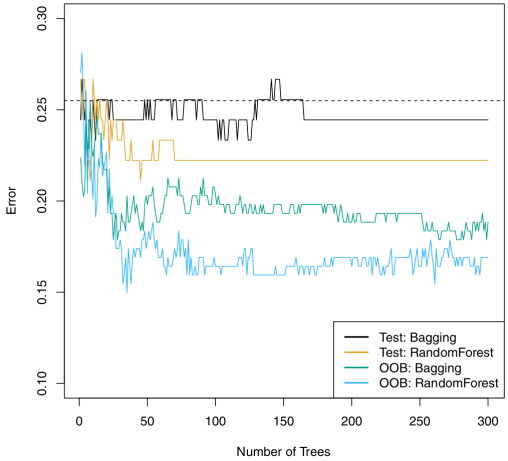
- *Blue is original data set* $X : n \times p$
- *Red are the subsampled data sets*
- *Build a tree from each*
- *Get prediction from each, return the average*

Prediction on new data point

New data point goes in, run through each tree, average outputs



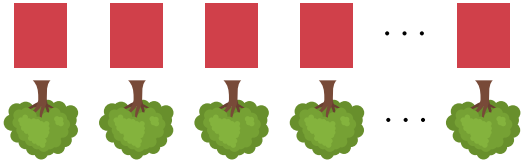
Example: Heart classification data



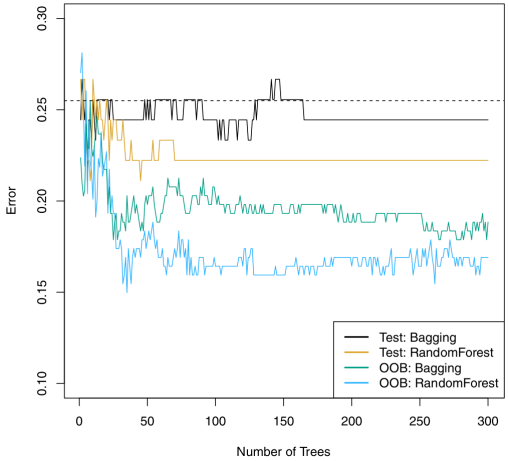
- Example is done for classification
- Ignore everything but black lines
- Dashed is test error from single classification tree
- Solid black is from bagged version
- Note that you just need B big enough for error to stabilize, so about 100 is good

Out of Bag Error Estimation

- On average, bootstrap sample uses about 2/3 of the data *We did this homework problem a while ago I think? Ch 5, Ex. 2*
- Remaining observations not used are called *out-of-bag* (OOB) observations
- For each observation, run through all the trees where it wasn't used for building
- Return the average (or majority vote) of those as test prediction



Error using OOB



- Green line has OOB error
- note OOB is averaged B/3 predictions per observation
- Book says the fact that the OOB value is so much lower is just random chance?

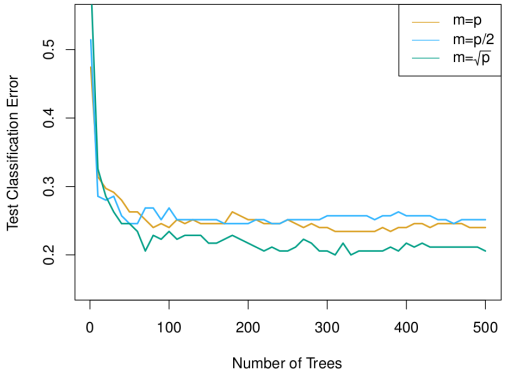
Section 3

Random Forests

The idea

- Goal is to decorrelate the bagged trees:
  - If there is a strong predictor, the first split of most trees will be the same
  - Most or all trees will be highly correlated
  - Averaging highly correlated quantities doesn't decrease variance as much as uncorrelated
- The random forest fix:
  - Each time a split is considered, only use a random subset of m the predictors
  - Fresh sample taken every time
  - Typically $m \approx \sqrt{p}$
  - On average, $(p - m)/p$ of splits won't consider strong predictor
  - $m = p$ gives back bagging

Example on gene expression



- Helpful for large p
- Predict one of 14 types of cancer (15 total levels), use 500 genes with most variance
- Random split to train and test set, three values of m
- Error rate of single tree: 45.7%
- Null rate 75.4% (classify each into the dominant overall class, here the healthy class)
- 400 trees enough for good performance
- $m = \sqrt{p}$ small improvement over bagging ($m = p$) in this case

Coding time!

TL:DR

- Bagging: trees grown independently on random samples. Trees tend to be similar to each other, can result in getting caught in local optima
- Random forest: trees independently on samples, but split is done using random subset of features

Next time

	F	10/17	***Review***		
	M	10/20	Fall Break		
	W	10/22	**Midterm #2**		
21	F	10/24	Polynomial & Step Functions	7.1-7.2	HW #5 Due Sun 10/28
22	M	10/27	Step Functions; Basis functions; Start Splines	7.2-7.4	
23	W	10/29	Regression Splines	7.4	
24	F	10/31	Decision Trees	8.1	HW #6 Due Sun 11/2
25	M	11/3	Random Forests	8.2.1, 8.2.2	
26	W	11/5	Maximal Margin Classifier	9.1	
27	F	11/7	SVC	9.2	HW #7 Due Sun 11/9
28	M	11/10	SVM	9.3, 9.4	
29	W	11/12	Single Layer NN	10.1	
30	F	11/13	Multi Layer NN	10.2	HW #8 Due Sun 11/16
31	M	11/17	CNN	10.3	
32	W	11/19	Unsupervised learning / clustering	12.1, 12.4	
33	F	11/21	Virtual: Project Office Hours		HW #9 Due Sun 11/23
	M	11/24	***Review***		
	W	11/26	**Midterm #3**		
	F	11/28	Thanksgiving		
	M	12/1	Virtual: Project Office Hours		
	W	12/3	Virtual: Project Office Hours		
	F	12/5			**Project Due**

Q of the day:
You have two very different datasets to create two very different models.
You have to use random forest on one and bagging on the other.
Which one would benefit more from random forest? what criteria would you use for the making the decision?